Water Security Toolbox: Problem Diagnosis Framework

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Overview: Water Security Toolbox

India is facing a water crisis due to over-exploitation, and climate change is exacerbating the issue. Numerous water security initiatives have been launched by Central and State governments, and corporate social responsibility programmes, to alleviate the looming crisis, but not all of these funds are spent in the most effective and efficient ways. Initiatives are often designed without diagnosing local hydrology and properly consulting local communities, and they often follow one-size-fits-all solutions that may be sub-optimally located. Impact assessments often measure success through simplified output indicators and do not provide enough learnings for improvement.

The draft Water Security Toolbox is developed by WELL Labs and the Environmental Defense Fund to address this gap. It offers a set of practical, science-based resources for designing and evaluating rural water security interventions, **built for field coordinators** in non-governmental organisations who need to make the best use of secondary data, conduct better fieldwork, and collect primary data in a structured way. The Toolbox comprises **three frameworks, two digital tools,** and **15 field-ready playbooks** for community leaders and community resource persons.

The Toolbox follows a sequential logic of diagnose, design and assess. The **Problem Diagnosis** component helps identify priority objectives for a watershed, drawing on the Water Index for Sustainability, Equity and Resilience (WISER), an integrated framework of six dimensions and 12 indicators that captures how communities experience water problems, such as groundwater decline or seasonal well drying. The **Solution Design** component matches validated objectives to landscape constraints to arrive at a shortlist of technically feasible interventions and a theory of change. The **Monitoring, Evaluation, and Learning (MEL)** component helps implement an outcome-led evaluation that measures water and socio-economic impacts, supported through simple planning, data collection and visualisation tools.

The broader vision for the Toolbox is to **democratise water science**, increasing the availability of **high-quality science at low cost and at scale** through standardisation of scientific processes and their delivery through open tools, increasing the agency of people to manage their own natural resources. It is envisioned as a **living ecosystem**, refined through a **prototype-implement-improve model**. Each deployment is treated as a cycle of use, review, and refinement, as more partners pilot it across diverse hydrogeological and institutional contexts.

Read more:

- A Primer on the Water Security Toolbox
- Solution Design Framework
- Monitoring, Evaluation, and Learning (MEL) Framework

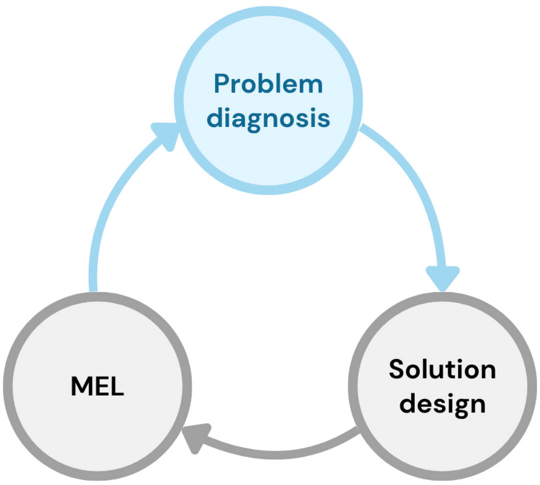
Use the tool

Visit the Water Security Toolbox website

Purpose of this Document

The objective of this document is to explain the logic behind the first component of the Water Security Toolbox – the Problem Diagnosis Framework – and walk readers through the digital tool. The Problem Diagnosis Framework helps those working on water security initiatives in rural Indian landscapes move from a broad water security concern to a clearer understanding of its causes. The tool serves as a platform that integrates secondary data, questions, hypotheses, and field observations into a single process. The document is intended for field coordinators and other practitioners planning water security work at the landscape level.

How Problem Diagnosis Fits in the Toolbox



The Water Security Toolbox follows a 'diagnose, design, and assess' sequence to solving rural water security challenges. **The Problem Diagnosis Framework is the first component, in which a user aims to understand the hydrological problems of a landscape and the issues experienced by local communities.** It is designed as a problem-first approach in which users are nudged to think through a problem, identify its exact root cause, and then develop solutions to address it.

The output of the Problem Diagnosis phase is a set of supported root causes and landscape objectives. These become the starting point for the Solutions Design Framework, where a user compares possible solutions to find those that work best for the landscape and its people. Then, the user moves on to the Monitoring, Evaluation, and Learning (MEL) Framework to assess whether the selected solutions met the objectives they were built for.

Need for a Structured Problem Diagnosis Process

Water-related work has been going on for decades and has delivered real gains. However, it has still not moved the needle far enough. The WHO/UNICEF assessment found that 2.1 billion people still lack access to safely managed drinking water (WHO & UNICEF, 2025). A World Bank review found that 30 to 40 per cent of rural water infrastructure in the countries it studied was either not functioning or performing below expected service levels (World Bank, 2017).

These numbers do not always point to a lack of effort or poor construction. A structure can be well designed and well built, but still be the wrong response for that area. An India-focused World Bank study found that poor siting, design, maintenance, and groundwater conditions can reduce the value of an intervention (James et al., 2015). A review of 86 studies reached a similar conclusion, showing that long-term performance also depends on local capacity, community involvement, and support after construction (Machado et al., 2022).

The challenge is that terms such as 'water scarcity' describe what people are experiencing but do not explain why it is happening. Two villages may both face water shortages after the monsoon for very different reasons. One may receive enough rainfall but quickly lose water from the landscape. Another may be extracting more groundwater than the aquifer can support.

This is where output targets can sometimes hide the mismatch between a solution and the problem. A programme can build thousands of structures and still find that farmers lack reliable water when they need it. A check dam, like any other intervention, is not automatically right or wrong. Its value depends on whether it responds to the main constraint in that area.

A structured Problem Diagnosis process creates a deliberate pause between identifying the concern and deciding on the solution. It helps ask questions like 'what is happening', 'where it is happening', 'why it is happening', and 'what evidence would help test that explanation'. Only after these questions are worked through do we define what needs to change, and then work towards making those changes.

What the Problem Diagnosis Framework Does

The Problem Diagnosis Framework can be considered a problem-first diagnostic thought process that is complemented by a digital tool. The workflow begins by reading the landscape using secondary data to observe peculiarities and ask 'why do they exist' and 'what needs to change'. These questions and hypotheses are then answered in the field and ultimately lead to the elucidation of the root causes of the water problems in the area. Once the root causes are known, solving the area's root problem becomes the objective, which is then addressed by the Solution Design Framework.

The Framework combines secondary data, hydrology, Water Index for Sustainability, Equity, and Resilience framework (WISER) indicators, field validation, and root-cause thinking. It does not replace field teams or hydrologists. It gives them a structured starting point so they can ask sharper questions, better document their observations, and use field time better.

Structure of the Approach

The Problem Diagnosis approach moves through five connected steps:

- Understand the landscape using secondary data
- Turn observations into hypotheses
- Verify hypotheses on the field
- Identify the root cause of problems
- Set the landscape objectives with the community

The steps are explained in a sequence below, but the process is not always one-way. Field evidence may show that an early assumption was weak or that an important layer was missed. This serves as a learning for future observations and addresses the problem of not knowing what we don't know. Thus, the purpose is to improve the diagnosis as the evidence becomes clearer with each completed workflow.

Step 1: Understand the Landscape Using Secondary Data

The Problem Diagnosis digital tool brings 19 scattered datasets together in one place, making it easier to read, compare, and form an initial view of the landscape. There are a number of secondary datasets available online, each that helps decipher a water problem in different ways. The Problem Diagnosis component curates 19 such datasets and brings them together for a select area. This saves time required to find and prepare data and reduces the risk of missing useful datasets because practitioners may not know they exist or how to use them.

For each dataset, the tool explains what it indicates, how it can be interpreted, and where its limitations need to be considered. This process helps examine the landscape from different angles and perspectives, reducing the risk of entering the field with a preferred/known solution or relying on a single dataset, which can introduce biases.

To understand the area, we need to begin with a unit boundary. The digital tool offers options to choose this boundary; the default is set to microwatersheds and watersheds as standard hydrological units. The unit is then used to clip all existing datasets, calculate statistics for each, provide interpretation guidance and prompts on what to check in the field.

Once all the layers are clipped to the unit boundary, they help answer questions regarding different parts of the water system. Elevation and drainage provide clues about how water may move through the terrain. Surface water, rainfall, soil, geology, and groundwater layers help us think about water availability, storage, and recharge. The land use land cover (LULC) and cropping intensity layers, and the canal networks shed light on the anthropogenic supply of water and its linkages to cultivation.

The digital tool also provides tips on how to interpret each layer, the most prominent category in each, what that might mean, and a short description of the resolution and how it might be limited or misleading. Depending on the statistics it generates based on that unit boundary, it also allows users to log relevant questions that they have in order to gain better clarity and context about the area.

Table 1: The layers used in the current tool.

Layer	What it Helps Us Understand
Watershed and Area of Interest (AOI) boundaries	Creates the shared unit for clipping, statistics, and comparison, using Hydrosheds Basin Level 12, Basin Level 7. It helps ask how conditions differ between the upper and lower parts of the watershed.
State boundaries	Places the selected watershed within a wider location map.
Digital Elevation Model (DEM)	Shows elevation, relief, and ridge-to-valley position, which help frame runoff, recharge, and erosion questions.
Drainage	Maps streams and nalas. It is also used to calculate drainage density. It helps understand how surface water moves through rivers and streams.
Land use land cover	Shows crop land, fallow and built-up lands, vegetation, and water bodies, giving clues about demand and land pressure.
JRC Water Occurrence	Shows how often surface water has been detected over the long term, from rare occurrences to permanent water.
JRC Water Transitions	Shows whether mapped water has remained, appeared, disappeared or shifted between seasonal and permanent states.
Cropping intensity	Shows repeated cultivation and provides clues about irrigation demand and access.
Canals	Shows surface-water delivery routes and how they relate to cropping patterns.
Command areas	Shows areas intended to receive canal water.
Lineaments	Shows mapped structural features that may influence groundwater movement.
Aquifers	Shows the broad aquifer setting and, where available, depth or yield information.
Climate trend	Provides long-term rainfall context for separating climate pressure from local supply, demand, or management issues.
Soil	Helps interpret infiltration, runoff, drought stress, and waterlogging.
Baseline demographics (population)	Combines village boundaries and population information to show who and where the field process should include.
Marginalised communities (% SC/ST), derived	Uses village demographic fields to flag where inclusion may need more attention.

Layer	What it Helps Us Understand
Groundwater stress (WISER)	Compares groundwater extraction to recharge and classifies the pressure level.
Irrigation access (WISER)	Uses long-term cropping intensity as a proxy for village-level irrigation access.
Kharif crop resilience (WISER)	Shows how the kharif-cropped areas hold up in deficient-rainfall years.
Rabi crop resilience (WISER)	Shows how the rabi-cropped areas hold up in deficient-rainfall years.

The idea is to look at the terrain using all the layers together while leaving notes on each layer individually. For example, the satellite image might suggest that the land is dry; the DEM may show that part of the watershed lies on a ridge or at a higher elevation with steep slopes; and the geology might indicate that the rock type is basaltic, a hard rock that could imply low groundwater storage.

At this stage, one is still making assumptions that need to be confirmed on the field. So, a user can log hypotheses like this onto a side panel in the tool: 'This area could have post-monsoon water scarcity due to the steep slopes and the low storage underground because of the hard rock aquifer'. Step 2 expands on this.

Step 2: Turn Observations into Hypotheses

Once the initial observations have been made, the next step is to convert them into questions that can be checked in the field. For instance, in another location within the same boundary as the above example, the land-use and cropping layers may show that most of the area is cultivated during the kharif season, while cultivation declines sharply during the rabi season. This pattern may point towards a possible water availability issue after the monsoon. However, the map does not explain why cultivation has declined. A user could then frame a question like 'Why does cultivation decline after the monsoon, and is water availability (or something else) the main reason?'.

The idea is to record what is observed, what remains unknown, and possible reasons for the patterns. This can be in the form of hypotheses or questions. Since these locations must be manually verified in the field, the digital tool allows users to mark observation zones around each location that require closer inspection and attach questions.

For the ridge example, the observation zone may include the ridge and a few locations slightly below it. The team may then ask when the wells begin to dry, whether they hold water after the monsoon, and whether wells lower on the slope behave differently. This helps the team determine whether water stress is related to topography, geology, demand, or a combination of these factors.

Writing the questions/hypotheses before fieldwork helps determine where the fieldwork needs to take place. It helps plan routes, coordinate group discussions, and conduct an Electronic Participatory Rural Appraisal (ePRA) with the villagers. This process is designed to add structure to fieldwork, develop assumptions, and coordinate with the required stakeholders.

Step 3: Verify Hypotheses on the Field

The purpose of fieldwork at this stage is to fine-tune the problem diagnosis to the local context. While secondary data helps us examine patterns across a large area, the information may be coarse, out of date, incomplete, or even misleading. For example, a dataset might show the presence of a canal, but the infrastructure might be dilapidated, preventing the water access it is intended to provide.

In this phase, the observation zones, questions, and hypotheses from the web app are packaged into a mobile app (QField) with a single click. A team now holds the entire curated dataset along with the users' questions, hypotheses, and observation zones on their mobile devices. The QField app also has the ability to record photographs, audio, and notes, which can then be linked to specific hypotheses to further support or refute them.

For the ridge example, the field team can now navigate to the ridge, check whether they are within the ridge observation zone they defined, and record data about the ridge. They can ask when water levels begin to fall, check well depths and local geology, look for signs of runoff and storage, and speak with users about how water availability changes throughout the year. If the area receives very little rainfall and the scarcity is primarily due to it, the hypothesis about steep slopes and low storage is refuted, and the root cause could be the low rainfall.

Field verification also needs to include community knowledge. This is facilitated through an Electronic Participatory Rural Appraisal (ePRA) session, during which a high-resolution satellite image of the area is used as a common reference. People are asked to discuss the water situation in different parts of the area, which sources are dependable, how conditions have changed, and what they think may help. This information provides details that the maps cannot show and may also reveal differences in access, priorities, and local experience. It goes hand-in-hand with the hydrological evidence when developing the diagnosis.

Step 4: Identify the Root Causes of Problems

After the fieldwork is complete, the team brings together the secondary data, field observations, and community knowledge to refine the diagnosis. The initial hypotheses are reviewed against the combined evidence to determine whether they still explain what is happening in the area and whether that explanation is sound.

The tool facilitates this with a simple click again. Within the mobile app, users can push changes back to the cloud, bringing all the data they collected in the field back to the web tool. Once the web tool is updated, it unlocks a new option to either validate, reject, or create new hypotheses.

Depending on the context gained, users are nudged to identify the root causes of a hypothesis. Building on the above example, if field observations confirm that the areas receive substantial monsoon rainfall on the ridge but retain very little water during the dry season, the hypothesis that post-monsoon scarcity is not related to rainfall holds true.

The next step is to question what exactly is the problem. So, now we look at all the layers together, and we know the rainfall is high, but the water does not stay, which means there is a problem with water retention. But why does the water not stay? Because the slopes are steep and the water is moving away, and the aquifer is a hard rock aquifer with limited to low storage. This forms the root cause of the problem.

Step 5: Set the Landscape Objective with the Community

Once the root causes have been identified, they can be converted into a broad objective for the area. **The objective is simply what needs to be addressed to solve this problem.** Users are not nudged to actually find solutions, but rather to define the outcome they want after the solutions are defined.

In the ridge example, if the main problem is low water retention and storage, the objective may be to increase local water retention and storage. These objectives are then handed over to the Solutions Design process, which determines the solutions that would actually help meet them.

What's Next?

Once the landscape objective is clear, the team moves to the second component of the Water Security Toolbox – the Solution Design Framework. At this stage, one can aim to identify water security solutions or interventions that best suit a landscape and the needs of people residing in it through a ground-up participatory process. Once the solutions are decided, the Monitoring, Evaluation, and Learning (MEL) component of the Toolbox comes into play.

Read more about the Solutions Design Framework

Acknowledgements

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This body of work has been shaped through sustained, ground-up engagement rather than a desk exercise. Over the past two years, the Toolbox has been informed by one high-level roundtable, seven on-the-ground test cases, and direct input from more than 100 stakeholders spanning civil society organisations, corporations, government functionaries, and farming communities.

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About WELL Labs

WELL Labs is transforming water systems at scale across India through research, partnerships, and collective action. It takes on audacious challenges, tackling complex problems where it believes it has a reasonable chance of success and the potential for large social returns. WELL Labs' work is science-led and community-focused, addressing the interconnections between water, the environment, land, and livelihoods. To create impact at scale, it embeds solutions within government systems, works with the private sector, and collaborates with civil society and active citizens.

WELL Labs is part of the Institute for Financial Management and Research (IFMR) Society, established in 1970 as a not-for-profit to provide research-based inputs to industries and the government. Together with Krea University and other centres at IFMR, WELL Labs is part of an ecosystem of labs and research centres with a mission to help prepare for an unpredictable world.

About EDF

A global non-profit, Environmental Defense Fund collaborates with governments, NGOs, research and academic institutions, corporates, and others to support and advance India's vision of shared, sustainable prosperity. EDF combines scientific and economic foundations, a broad network of partnerships and a pragmatic approach in support of India's ambitions. Its areas of interest include demonstrating the viability of sustainable livelihoods in agriculture, ensuring sustainable groundwater supplies for agriculture, livestock and fisheries, establishing the shareholder value potential through responsible business, informing of the potential of market-based mechanisms, and catalysing the climate technology ecosystem in India.

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